ONE-PAGE BRIEF

Post-quantum cryptography review: foundry @ 7f0a409a5c20

Client: Foundry (Paradigm) — public repository (passive review) · Source: github.com/foundry-rs/foundry, commit `7f0a409a5c20` · Reviewed: 2026-11-04 18:46 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 4 of 4

post-quantum by default in the critical paths

FINDINGS

0 Critical **4** High **7** Medium **5** Low **3** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Keep the inventory current and review new code paths before release.

WHAT MATTERS MOST

High QSI-2026-11922	**Possible hard-coded secret in crates/evm/symbolic/src/runtime/solver/mod.rs** preliminary What to do: Move secrets to a secrets manager/environment; rotate the exposed values.
High QSI-2026-11924	**Non-cryptographic generator next to key or seed generation in crates/anvil/src/cmd.rs** preliminary What to do: Use the platform CSPRNG (secrets, os.urandom, crypto.getRandomValues, crypto/rand, SecureRandom, OsRng) or a hardware TRNG for every key...
High QSI-2026-11930	**quick-xml 0.39.4 is affected by RUSTSEC-2026-0194** What to do: Upgrade to a fixed version (0.41.0).
High QSI-2026-11931	**quick-xml 0.39.4 is affected by RUSTSEC-2026-0195** What to do: Upgrade to a fixed version (0.41.0).

12 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium channel key exchange (TLS, Noise, handshakes)	TLS ECDHE	Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols)
Medium transaction and message signing	ECDSA, BLS12-381, Ed25519	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium authentication and identity (certificates, tokens, device attestation)	ECDSA	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them
Medium role not determined	BLS12-381, ECDSA	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 6; unlikely to apply to this build: 1 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 5.

Static review of source code and manifests; files analysed: 760. No requests were sent to the owner's systems.

Full report, content SHA-256: `11eb453dbcef9b595c5e7749f4a83429e0cc136eb0a9ec83e490d1e351eb49d4`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.